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Investigating the behavioural responses of trapped fishes using underwater video surveillance

G. F. Renchen*†‡, S. J. Pittman* and M. E. Brandt†

**Biogeography Branch, National Oceanic and Atmospheric Administration, 1305 East-West Hwy, 9th Floor, Silver Spring, MD 20910, U.S.A. and †Center for Marine and Environmental Studies, University of the Virgin Islands, #2 John Brewers Bay, St Thomas, VI 00802, U.S.A.*

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An underwater video surveillance system known as TrapCam was used to continuously record (15 × *c*. 24 h periods) fish behaviour within and immediately surrounding an experimental fish trap situated in a coral reef ecosystem in the United States Virgin Islands. Of the 100 fishes (18 species, 12 families) trapped, surgeonfishes (Acanthuridae) and snappers (Lutjanidae) were most common. Thirteen distinctively identifiable behaviours were observed for trapped fishes. Species did not differ significantly in the proportion of time allocated to different behaviours (ANOSIM, $R = 0{\cdot}142$). Doctorfish *Acanthurus chirurgus* and grey angelfish *Pomacanthus arcuatus* allocated the largest proportion of their recorded time to enter and exit the trap. Fishes spent an average of 15 min in the trap before escaping. Sixty-seven per cent of trap approaches consisted of an individual of the same species as one already trapped suggesting that conspecific attraction may have occurred. Fifteen per cent of trapped species were observed with abrasions to the head and 70% were observed approaching the trap corners. The results of this study provide a greater understanding of the behavioural interactions between fishes and traps that can help explain patterns of catch composition, the physical condition of fishes in traps and inform design of gear modifications to optimize by-catch reduction in the trap fishery.
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Key words: Caribbean; conspecific attraction; fish traps; ghostfishing; marine debris; U.S. Virgin Islands.

INTRODUCTION

Fish traps are an economically important gear used throughout the Caribbean to catch a variety of reef fishes and crustaceans (Munro, 1974; Garrison *et al*., 2004; Shärer *et al*., 2004; Marshak *et al*., 2008). Fish traps may become lost or abandoned (referred to as derelict) due to processes such as inclement weather, entanglement with benthic substrates, intentional abandonment of old and damaged traps, loss of trap markers coming into contact with boat propellers and other gears, and human error (Laist, 1995). Derelict fish traps have the potential to ghostfish, the process

‡Author to whom correspondence should be addressed at present address: Center for Marine and Environmental Studies, University of the Virgin Islands, #2 John Brewers Bay, St Thomas, VI 00802, U.S.A. Tel.: +1 3302776697; email: gabby.folger@gmail.com

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by which derelict fishing gear catches marine fauna and induces mortality in an uncontrolled manner (Matsuoka *et al*., 2005; Smolowitz, 1978). Direct observations of the type and duration of behaviours exhibited by fishes while in and around traps provides a greater understanding of the ecological function of fish traps and of the influence of traps on the dynamics of catch composition as well as the physical condition of target species and by-catch. Such information can provide new insights on fish–trap interactions with potential to inform managers and fishermen interested in ensuring that fisheries minimize impacts from gears and utilize sustainable practices such as designing gears for by-catch reduction. Attempts to modify gears or improve management efforts may prove difficult if there is insufficient scientific knowledge and understanding of the behavioural responses of fishes to gear (Bardach & Magnuson, 1980).

Studies of fish traps have assessed the mode of operation and selectivity of traps (Munro *et al*., 1971; Gobert, 1998) and impacts to coral reef ecosystems (Sheridan *et al*., 2006; Hawkins *et al*., 2007); but very few studies have experimentally addressed the phenomenon of ghostfishing and the behavioural responses of fishes to traps. *In situ* studies with observers using scuba indicated that fishes inside traps behaved abnormally, including repeatedly butting into the trap mesh, often resulting in injury (Luckhurst & Ward, 1987). While diver observations can provide insightful data, the operations can be expensive, are often of relatively short duration and may introduce inaccurate and biased behavioural results due to the presence of a human observer (Jury *et al*., 2001). Remote underwater video observations provide behavioural information, with minimal bias to normal behavioural patterns (Barnes, 1963; Jury *et al*., 2001; Pratt *et al*., 2005), yet such remote observations have not been conducted for Caribbean trap fisheries.

Information on fish behavioural interactions with fish traps leads to a more complete understanding of the behavioural processes involved in trapping fishes and can provide information on the design of gears that minimizes unintended by-catch as well as injuries and mortality of trapped fishes. Behavioural characteristics and responses of fishes to gear have been used to develop by-catch reduction devices in commercial trawling industries (Broadhurst, 2000; Olla *et al*., 2000) and evaluation of these devices has been improved through the use of underwater video (Workman & Foster, 1994). The susceptibility to capture, interactions between species and relationships between the size and behaviour of trapped fishes are all ecologically relevant trap–fauna relationships that need to be considered for effective management.

This study focused on the trap fishery of the U.S. Virgin Islands, which is mostly concentrated in the waters surrounding the northern islands of St Thomas and St John. The trap fishery in the U.S. Virgin Islands accounts for *c*. 37% of the total landings and revenue making it the largest economic component of the local multi-gear, multispecies fishery (Agar *et al*., 2005). The rectangular and chevron traps are the most commonly used designs, typically constructed of steel rebar or wood. Approximately 3662 fish and 2446 lobster traps are fished between St Thomas and St John with 100% of the produce sold for local consumption by residents and tourists (T. Blanchard, pers. comm.). The objectives of this study were to identify and characterize fish behaviour when inside and approaching a fish trap, develop a method to quantify fish–trap interactions *in situ*, categorize and quantify the allocation of time to different behaviours and determine if temporal patterns exist in the entering and exiting of traps by fishes. It was expected that fishes would have a temporal pattern of

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capture scheduled with known periods of foraging activity and that species-specific differences would exist in the proportion of time allocated to different behaviours while trapped.

MATERIALS AND METHODS

TRAP DESIGN

A rectangular fish trap design commonly utilized by fishermen of the U.S. Virgin Islands was constructed by a local fisherman to meet fishing regulations required by the U.S. Virgin Islands Department of Planning and Natural Resources (United States Virgin Islands Department of Planning and Natural Resources, 2009). The trap consisted of a steel rebar frame, horseneck funnel and 50·8 mm wire-coated square mesh (Fig. 1). The escape panel on the side of the trap was tied closed with 3·2 mm diameter untreated jute twine.

TRAPCAM SYSTEM

A custom-built underwater video surveillance system, referred to as TrapCam, was used to collect underwater video footage of fish behaviour. The system consisted of a 540TV line resolution bullet camera with an automatic switch from colour to black and white video under low-light conditions (0·05 lx, 4 mm lens, Model 214157-01; www.supercircuit.com). A solid state compact flash digital video recorder (DVR; www.chasecam.com) powered by a 12 V, 20 Ah Nimh battery connected to the camera was required to continuously record in high resolution for 24 h. Video files were stored in 4 h segments on a 32 GB compact

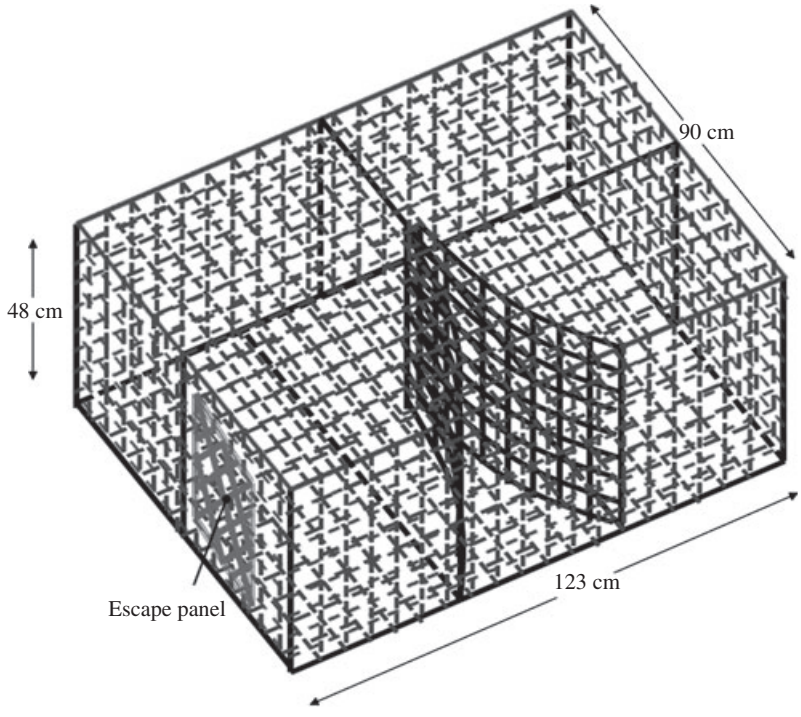

FIG. 1. Diagram of the rectangular trap used in the TrapCam study.

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flash memory card within the DVR. The entire system was mounted in a cylindrical PVC underwater housing pressure tested to *c*. 40 m (130 feet) (Bellamare Subsea Engineering and Scientific Imaging; www.bellamare-us.com). Due to the nature of the PVC housing, the TrapCam unit had to be turned on, set to record and then locked in the housing on dry land prior to each video sample. For the same reason, video footage could not be downloaded or analysed until the TrapCam was returned to dry land.

The TrapCam was made negatively buoyant with lead weights and mounted on the seafloor on a concrete block *c*. 1 m from the entrance funnel of the fish trap. This provided a field of view large enough to encompass the entire trap (*c*. 1·23 m wide) as well as *c*. 1 m above and to either side of the trap. Two Princeton Tec Shockwave LED (Princeton Tec; http://princetontec.com/?q=shockwave-led/) underwater dive lights that provided a maximum of 20 h of light were attached to stakes and mounted at the two opposing rear corners of the trap, with beams aimed at the entrance funnel. The lights were covered with a magenta coloured filter (Rosculux #46 Magenta, 6 and 85% transmission at 440 and 740 nm wavelengths, respectively, Rosco Laboratories; www.rosco.com) to decrease the influence of artificial light on fish behaviour. Filter selection was based on information published by Stoner *et al*. (2008) and Widdler *et al*. (2005) whom suggested that artificial light in the range of 400–700 nm wavelengths tended to influence fish behaviour and that artificial illumination in the far red portion of the spectrum (≥680 nm wavelength) would be less obtrusive than white light.

Prior to the beginning of this study, six pilot samples were conducted with the TrapCam system; three samples with underwater lighting and three without to determine the recording capabilities of the camera, optimize the position of lights and camera and determine if fishes were attracted to the lights. Each video sample in the pilot study captured a combination of day- and night-time recordings and did not reveal any attraction of fishes to the lights based on the absence of fishes moving towards or aggregating or avoiding the beams of magenta light.

STUDY SITE AND FIELD SAMPLING

The fish trap and TrapCam were deployed at Range Cay (18·339444°N; 64·9878451°W), a shallow (5 m depth) nearshore hard bottom area with scattered coral communities in St Thomas, U.S. Virgin Islands, during August 2010 to March 2011. Site selection was determined by the presence of active and derelict fish traps, recreational and or unregistered fishers and its proximity to the University of the Virgin Islands' Center for Marine and Environmental Studies in Brewer's Bay, St Thomas. Observations and conversations with local artisanal fishermen confirmed Range Cay was regularly fished.

The fish trap was deployed unbaited as is common practice in the U.S. Virgin Islands (D. Olson, pers. comm.) and left in the same location for the duration of the study to simulate a derelict trap. Video samples were only collected when fishes were present in the trap. Sample collection was contingent upon fishes being caught in the trap, weather conditions, water visibility and the TrapCam functioning properly. Continuous 24 h video recordings of a single un-baited rectangular fish trap were initiated between 1500 and 1600 hours prior to the first twilight period. After each 24 h video sample, only the TrapCam and lights were returned to land by scuba divers where the video footage was downloaded, batteries were charged and the TrapCam system was allowed to cool. All trapped fishes were released after each sample and the trap was left closed, actively fishing between observation periods. Two usable video samples were collected each month *c*. 3–7 days apart.

Fifteen samples were collected totalling 326·3 h of recorded observation, each of which was time-stamped. The aim was to collect continuous samples over 24 h; however, due to equipment malfunctions (*i.e.* camera overheating, batteries losing charge and lights flooding), the duration of sample recording times ranged from 15·08 to 24·86 h with an average of 21·75 h. In addition, the TrapCam battery system was only capable of recording for *c*. 24 h; therefore, a recording to match the typical soak of commercial trap fisheries (7 days) was not possible. Diel observations, however, were considered sufficient to capture changes in behaviour from day to night.

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VIDEO ANALYSIS

Video samples were analysed for catch composition, fish entry and exit times, activity periods and behavioural interactions of fishes both inside and immediately outside the trap. Trapped fishes were defined as those retained in the trap that were too large to escape through the trap mesh (*i.e.* they could only escape through the trap entrance funnel). Trapped fishes were classified as commercial or by-catch species based on results from a by-catch study (Olsen, 2008) and anecdotal information obtained from the St Thomas Fishermen's Association. Olsen (2008) identified by-catch species for the St Thomas trap fishery based on food preferences, size and risk of ciguatera poisoning. Video footage was continuously recorded once the TrapCam was turned on therefore post-retrieval video analysis did not begin until *c*. 10 min after divers were observed leaving the study site and disturbed fishes were allowed to acclimate. The start and end time of fish activity periods were recorded and used to determine what proportion of an entrapped fish's time was allocated to particular activities during the daylight hours only. Start and end times were recorded when a significant change in activity occurred. The ability to rewind the recording allowed the duration of the activities to be calculated. In addition, to examine frequency of occurrence for any particular activity type, counts were made of the number of times a behavioural interaction occurred. For instance, if a fish displayed aggression towards another species this was documented as a count rather than an amount of time.

An ethogram was developed to classify the distinctive types of fish behaviours (Supporting Information Table SI). Individual fishes were identified through visual total length (L_T) estimation, using the known trap mesh dimensions (50·8 mm mesh) as a size reference. When multiple fish of the same species were trapped, it was determined through trial and error that the behaviour of only two individuals could be accurately followed at one time. In an effort to avoid inflating the catch information, individual fishes observed entering and exiting multiple times while staying in the camera field of view were only counted once for the total catch. Fishes that approached the traps or were visible within 1 m of the traps were also recorded. Due to a narrow beam width from the underwater lights, behaviour at night was not discernible within the trap beyond the entrance funnel; therefore, night observations were limited to recording of entry and exit times.

DATA ANALYSES

The six most frequently observed behaviours were: (1) entering and exiting, (2) grazing, (3) an individual approaching mesh, (4) aggregating with other fishes while approaching mesh, (5) staying in one place and (6) other. The category other, included all other less frequently observed behaviours used in the analyses. Only species where four or more individuals were identified throughout the samples were included in the behavioural allocation analyses. The amount of time a fish allocated to each behaviour was converted into a proportion. During the analyses, each fish was assigned to its own row, representing only one sample. The different behaviours for each sample may be somewhat dependent upon each other, but multivariate statistics including those used here were developed to deal with potentially correlative datasets. Data were square root transformed and a similarity matrix calculated using Bray–Curtis similarity coefficients (Clarke, 1993). The fish samples were analysed for differences among individual fish species using the multivariate ordination technique of non-metric multidimensional scaling (nMDS) applied in PRIMER-e software (Clarke & Gorley, 2006). Within the nMDS ordination plot, fishes exhibiting similar behavioural allocation times are closer together in ordination space (Clarke, 1993). The quality of the nMDS ordination was determined based on Kruskal's stress which is a measure of the match between calculated similarities and distances between points in the ordination. Lower stress values indicated a lower risk of misinterpretation of the ordination: stress levels $<0{\cdot}3$ indicated a good match and stress values $<0{\cdot}1$ were ideal (Fasham, 1977; Clarke & Warwick, 2001).

Statistical tests of differences in behavioural allocation among fish species were also performed based on the Bray–Curtis similarity matrix among samples by applying the technique analysis of similarities (ANOSIM), a multivariate analogue of ANOVA (Clarke, 1993; Clarke & Warwick, 2001). The null hypothesis that there was no significant difference among groups was rejected when the significance level (P) was $<0{\cdot}05$; however, the extent of

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any significant difference was determined in the context of the more informative R-statistic and visual interpretation of nMDS ordination plots. The R-statistic value typically falls between 0 and 1, indicating some degree of discrimination between samples. Clarke & Warwick (2001) interpretations of ANOSIM R were followed as: $R < 0{\cdot}25$ = barely separable; $R > 0{\cdot}5$ = overlapping, but clearly different and $R > 0{\cdot}75$ = well separated. The number of fishes outside that approached the trap was also recorded. A simple linear regression was used to determine if fishes were more likely to approach the trap when more fishes were trapped inside using the PASW v. 18 statistical software programme (SPSS Inc.; www.lbm.com/spss_statistics).

RESULTS

CATCH COMPOSITION

In the 326·3 h of footage, 100 fishes were identified within the trap comprising 12 families and 18 species (Fig. 2). The most frequently observed families were surgeonfishes (*Acanthuridae*, $n = 29$ individuals), snappers (*Lutjanidae*, $n = 20$), boxfishes (*Ostraciidae*, $n = 15$), angelfishes (*Pomacanthidae*, $n = 8$) and porgies (*Sparidae*, $n = 8$). The most frequently observed species were: doctorfish *Acanthurus chirurgus* (Bloch 1787), $n = 28$ individuals, smooth trunkfish *Lactophrys triqueter* (L. 1758), $n = 16$, yellowtail snapper *Ocyurus chrysurus* (Bloch 1791), $n = 11$, saucereye porgy *Calamus calamus* (Valenciennes 1830), $n = 8$ and grey angelfish

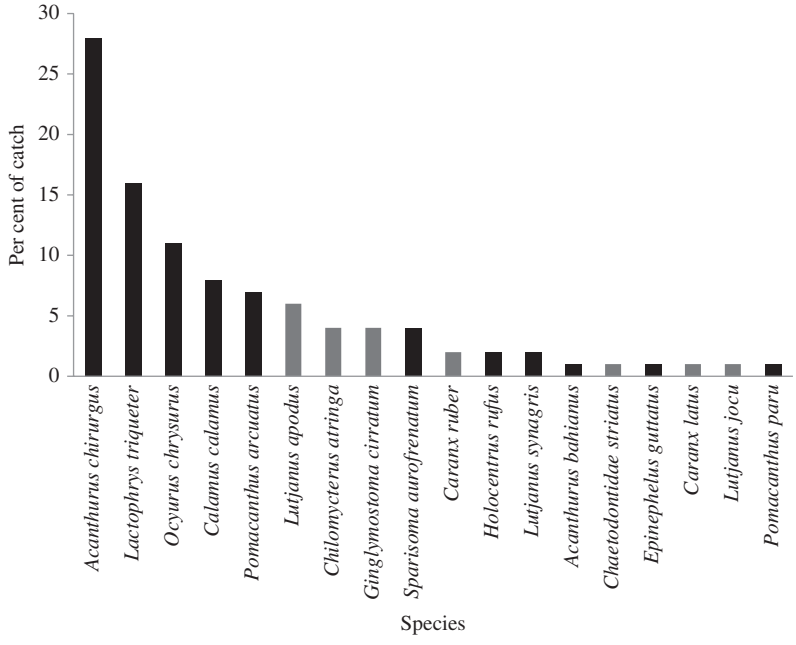

FIG. 2. Proportion of the total catch (abundance) in the fish trap grouped by fish species for all 15 video samples (326·3 h): Commercial (■) or by-catch (■) species represent their importance to the commercial trap fishery in St Thomas, U.S. Virgin Islands.

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TABLE I. The mean ± S.E. per cent of time, species with four or more samples, allocated to the six most frequently observed behaviours

Species	EE	G	IAM	SIOP	SSCAM	Other
Acanthurus chirurgus	44·3 ± 9·3	26·1 ± 7·3	19·8 ± 7·5	1·9 ± 1·9	7·7 ± 5·1	0·2 ± 0·2
Calamus calamus	—	—	24·2 ± 11·6	14·3 ± 7·7	61·5 ± 18·3	—
Chilomycterus atringa	—	—	2·0 ± 1·1	32·8 ± 11·4	65·1 ± 11·9	0·1 ± 0·1
Ginglymostoma cirratum	—	—	43·8 ± 6·8	55·8 ± 6·5	—	0·4 ± 0·4
Lactophrys triqueter	0·1 ± 0·2	8·6 ± 5·3	60·6 ± 11·0	0·8 ± 0·8	29·8 ± 11·5	0·1 ± 0·1
Lutjanus apodus	—	—	97·3 ± 2·0	2·7 ± 2·0	—	—
Ocyurus chrysurus	—	—	32·7 ± 11·9	1·0 ± 0·7	64·8 ± 13·1	1·5 ± 1·5
Pomacanthus arcuatus	32·5 ± 20·6	9·3 ± 9·3	12·5 ± 7·9	—	45·7 ± 22·8	—
Sparisoma aurofrenatum	0·3 ± 0·3	—	95·5 ± 4·1	4·2 ± 4·2	—	—

EE, entering and exiting; G, grazing; IAM, individual approaching mesh; SIOP, staying in one place; SSCAM, schooling, scanning, approaching mesh; other, all other less frequently observed behaviours.

Pomacanthus arcuatus (L. 1758), $n = 7$. The minimum and maximum L_T recorded for trapped fishes were 12 and 100 cm, with a mean ± S.E. L_T of 25·2 ± 1·8 cm. The smallest individuals caught were a striped butterflyfish *Chaetodon striatus* L. 1758, 12 cm L_T and *L. triqueter*, 12 cm L_T, and the largest was a nurse shark *Ginglymostoma cirratum* (Bonnaterre 1788), 100 cm L_T. Two fish carcasses were observed in the traps (*C. calamus* and *L. triqueter*), however, the moment of death was not witnessed.

FISH BEHAVIOUR

A wide variety of fish behavioural interactions were observed in the video samples, with 13 behaviours expressed among individual species and families (short sections of actual video footage are provided in Movies SI and SII, Supporting Information). The proportion of time allocated to the top six behaviours was not significantly different among fish species (ANOSIM Global $R = 0{\cdot}142$, $P < 0{\cdot}01$) (Table I) which was supported by the nMDS ordination plot that revealed high overlap in the behaviours exhibited by all fishes. A low stress value of 0·05 suggested that the nMDS ordination (Fig. 3) provided a nearly perfect two-dimensional representation of the multivariate behavioural data therefore allowing for a reasonable interpretation. While there was not a significant difference in the proportion of time allocated to the different behaviours among species, the amount of time allocated to individual behaviours was not equal.

Fishes were also observed entering and exiting (escaping) the trap through the entrance funnel. A total of 167 entrances and 175 exits were recorded (Fig. 4). The number of entrances and exits were not always equal due to fishes observed entering but not exiting prior to the end of the sampling period or fishes present in the trap at the beginning of the sampling period were not observed entering, but perhaps exiting. There were more entrances and exits than the total number of fishes caught because several of the fishes moving into and out from the trap were the same individuals observed on multiple occasions, that never left the camera field of view. Fish species spent 58·04 ± 15·75 min (mean ± S.E.) in the trap before escaping through the entrance funnel. Ten species were observed exiting the trap with

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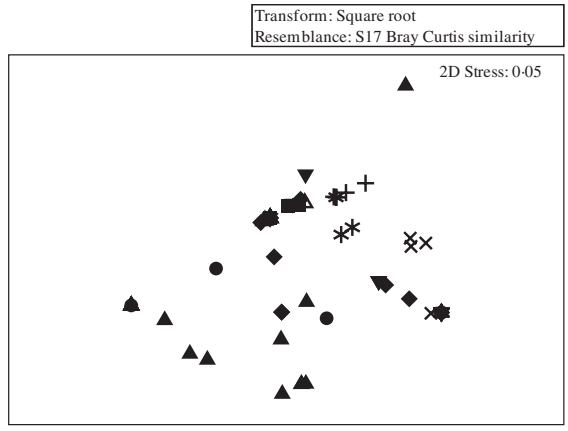

FIG. 3. Non-metric multidimensional scaling ordination of the proportion of time allocated to the top six behaviours by individual fishes with an abundance greater than or equal to four (▲, *Acanthurus chirurgus*; ▼, *Ocyurus chrysurus*; ■, *Lutjanus apodus*; ◆, *Lactophrys triqueter*; ●, *Pomacanthus arcuatus*; +, *Ginglymostoma cirratum*; ×, *Chilomycterus atringa*; ∗, *Calamus calamus*; △, *Sparisoma aurofrenatum*).

A. chirurgus, the longspine squirrelfish *Holocentrus rufus* (Walbaum 1792), *L. triqueter* and *P. arcuatus* comprising *c*. 66, 17, 6 and 6% of the escapes (Fig. 5). The schoolmaster *Lutjanus apodus* (Walbaum 1792), *C. calamus* and lane snapper *Lutjanus synagris* (L. 1758) each accounted for *c*. 1% of the escapes, while *C. striatus*,

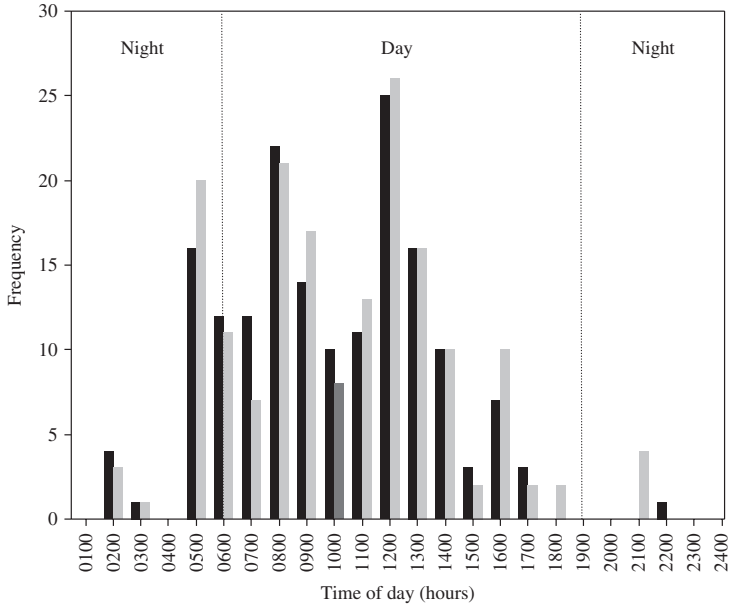

FIG. 4. Frequency distribution of the number of exits (■) and entrances (■) from the fish trap recorded per hour in a day for all video samples.

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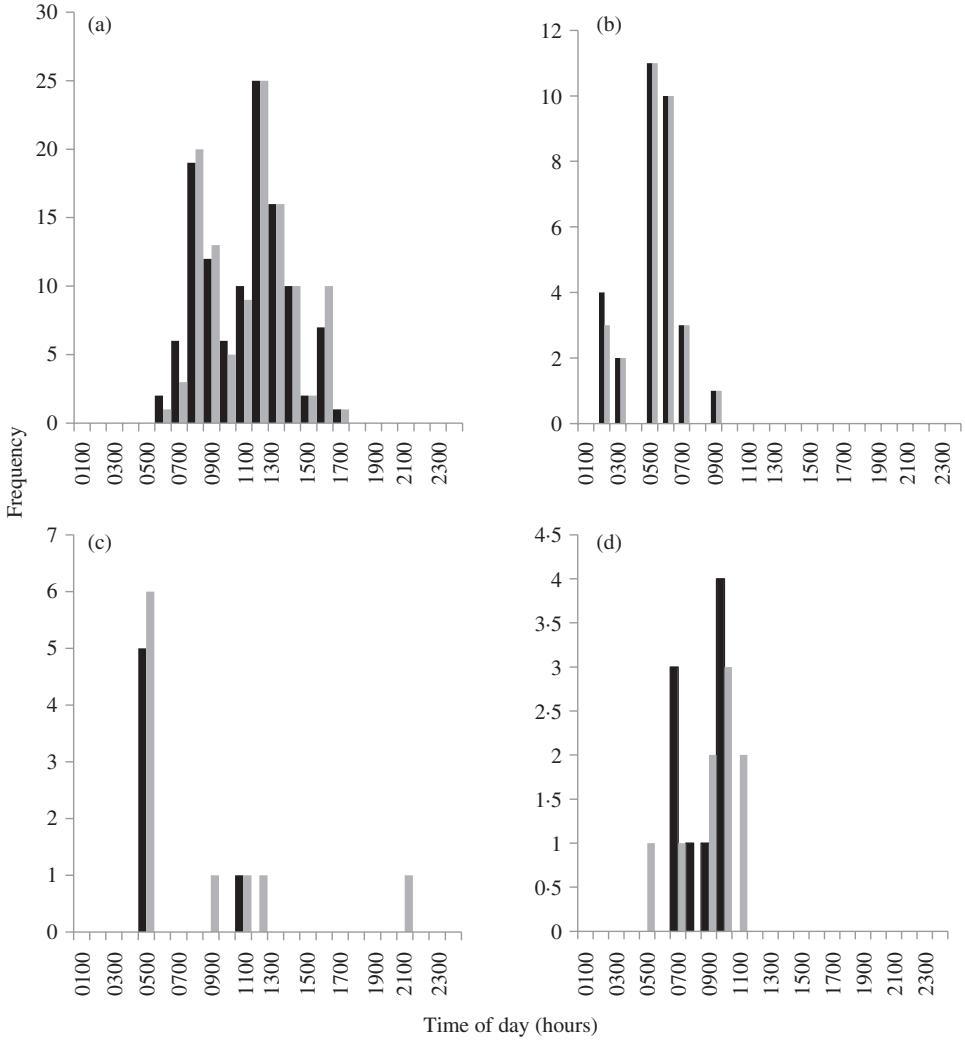

FIG. 5. Species-specific frequency distributions of the number of exits (■) and entrances (■) from the fish trap recorded per hour in a day for all video samples. The species presented, (a) *Acanthurus chirurgus,* (b) *Holocentrus rufus,* (c) *Lactophrys triqueter* and (d) *Pomacanthus arcuatus*, were observed entering and or exiting the trap on four or more occasions.

red hind *Epinephelus guttatus* (L. 1758) and redband parrotfish *Sparisoma aurofrenatum* (Valenciennes 1840) each accounted for 0·6% of the escapes. Seventy trapped fishes were observed approaching the mesh at the corners of the trap at least once. Species most often observed banging into the trap mesh included *O. chrysurus*, *C. calamus* and *L. triqueter*. Fifteen per cent of trapped fishes were observed with abrasions, concentrated mostly near their snouts. Injured species included *O. chrysurus* (67% of the injuries), *C. calamus* (13·3%), bar jack *Caranx ruber* (Bloch 1793) (6·7%), *S. aurofrenatum* (6·7%) and dog snapper *Lutjanus jocu* (Bloch & Schnedier 1801) (6·7%).

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SPECIES OUTSIDE THE TRAP

Twenty-two fishes outside were observed approaching the trap entrance funnel when other fishes were inside, while only six were observed approaching when no other fishes were inside the trap. Eighteen per cent of the approaches resulted in entrapment. The presence of fishes inside the trap appeared to attract other fishes to the trap, often those of the same species; a simple linear regression revealed that the number of fishes present in the trap was not a useful predictor of the number of approaches ($R = 0{\cdot}142$, $F_{1,10}$, $P > 0{\cdot}05$, 95% C.I. $-0{\cdot}28$, $1{\cdot}05$). Approximately 67% of the fishes that approached the trap entrance funnel were conspecifics of those already trapped. Seven fishes outside the traps were observed swimming alongside a conspecific on the inside of the trap. Species exhibiting conspecific attraction were *A. chirurgus*, *L. triqueter*, spotted burrfish *Chilomycterus atringa* (L. 1758), *P. arcuatus* and *C. calamus.*

Attraction of predators to the trap was also observed. Predatory fishes identified included cubera snapper *Lutjanus cyanopterus* (Cuvier 1828), $n = 17$, barracuda *Sphyraena barracuda* (Edwards 1771), $n = 11$, tarpon *Megalops atlanticus* Valenciennes 1847, $n = 7$, crevalle jack *Caranx hippos* (L. 1766), $n = 5$, bar jack *Caranx ruber* (Bloch 1793), $n = 2$ and horse eye jack *Caranx latus* Agassiz 1831, $n = 2$. Of the predatory fishes observed, three species (*L. cyantopterus*, *C. hippos* and *S. barracuda*) attempted to enter the traps or reach fishes through the mesh. The *L. cyanopterus* was observed circling the trap on several occasions, while three *C. hippos* were observed on one occasion, both of which caused fishes inside the trap to move to the opposite side of the trap. The other fishes were observed frequently swimming over the top of the trap or directly in front of the camera and trap entrance.

DISCUSSION

Due to the passive nature of fish traps, catch is dependent on fishes actually encountering the gear and their vulnerability to entrapment within the gear at the time of gear retrieval (*i.e.* entering and not being able to escape) (Robichaud *et al.*, 2000; Stoner, 2004). Studies have suggested that the physical structure of traps and the presence of predators and prey can also influence whether fishes enter traps (High & Beardsley, 1970; Matthews, 1985; G. Renchen, S. Pittman, R. Clark, C. Caldow, D. Olsen & R. Hill, unpubl. data). The results of this study demonstrated that species-specific behavioural responses to traps and trapped fishes (*i.e.* presence and absence of conspecifics) may also influence whether fishes enter, remain in or escape from traps.

A mixture of coral reef-associated species was caught in this study that included diurnally and nocturnally active species representing a wide range of herbivores, invertivores and piscivores. The composition of catch in this study was similar to that in other trap catch studies conducted in the U.S. Virgin Islands (Garrison *et al.*, 2004; Trumble *et al.*, 2006; Olsen, 2008); however, not all habitats or depths utilized by commercial and illegal trap fishermen could be investigated in this study. The present results may not be entirely representative of fish catch or behaviour in all the various habitat types across the shelf, but they do provide useful information for guiding future work of trap-related fisheries and the application of underwater video techniques for behavioural studies in marine fisheries ecology.

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Fishes were observed expressing multiple behaviours while trapped; however, there was unexpected overlap among species in the proportion of time allocated to these behaviours. The near zero *R* statistic of the ANOSIM suggested that a distinguishable pattern in behavioural allocation among species was not present, but the nMDS ordination suggested that *A. chirurgus* and *P. acruatus* may be different as they appeared to form a grouping in the lower left portion of the ordination plot. Both these species allocated the greatest amount of their time to entering and exiting the trap.

Entrances and escapes of fishes peaked in the morning and then tapered off after 1200 hours. This may be attributed to the natural diurnal and nocturnal activity patterns exhibited by fishes (Helfman, 1993). Although behaviour from the night-time footage was not analysed due to the inability to light the entire trap, trapped nocturnal species such as snappers, jacks and nurse sharks could be seen moving within the trap. In contrast, diurnal species such as *A. chirurgus* were repeatedly observed entering and exiting the trap during the daylight hours only. Similar behaviour has been documented, as *A. chirurgus* comprised the largest percentage of the catch and had relatively short residence times (<6 days) with no observed mortality (G. Renchen, S. Pittman, R. Clark, C. Caldow, D. Olsen & R. Hill, unpubl. data). The relative mobility, body form and habitat adaptations of individual fishes and species of fishes may also determine susceptibility to entrapment. More mobile species such as surgeonfishes that are accustomed to navigating structurally complex reefs may be more likely to encounter and enter a trap, but high activity rates and maneuverability may increase the likelihood of exiting a trap (Robichaud *et al.*, 2000). *Acanthurus chirurgus* and *P. arcuatus* are considered to be very manoeuverable species that utilize short, straight bursts and relatively low turning radii while swimming and navigating complex habitats (Domenici & Blake, 1997; Webb & Fairchild, 2001). *Lactophrys triqueter* also manoeuvres reefs, but it does not have a highly manoeuverable body shape, perhaps owing to its lower rate of escapement. Functional ecology studies have suggested that morphological differences may be correlated with behavioural differences (Gerstner, 1999), and in the present study, *A. chirurgus* and *P. arcuatus* exhibited behaviours slightly different from the other species as was evidenced by the hint of separation of these two species from other species in the nMDS ordination and the large number of escapes. Less mobile and large bodied elongated species such as *G. cirratum*, which allocated the majority of time in traps lying relatively sedentary, may be less likely to find the exit.

Physical disturbances can disrupt or alter the normal behavioural and movement patterns of fishes, leading to a state of physical and neurological distress (David & Closs, 2001). Most fishing techniques are recognized to induce stress responses in fishes due to the high amount of energy expended and subsequent fatigue caused by attempting to escape (Skomal, 2007). Fishes expending large amounts of energy attempting to escape could detract from foraging and hasten starvation depending on several factors such as the amount of food in the stomach when trapped, and the metabolic debt and rate of oxygen consumption (Weber & Haman, 1996; Stoner, 2004; Brown *et al.*, 2005). In this study, species did not exhibit a significant difference in the proportion of time allocated to different behaviours in response to entrapment; however, differences among species may exist in the way in which these behaviours translate to physiological stress impacts.

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Fishes suffering from predation attacks or mechanical skin damage may emit chemical cues, also known as alarm responses (Golub *et al*., 2005). Responses of fishes to alarm responses vary, but those that are confined in a small area or are wary of becoming trapped may react more strongly to alarm responses (Magurran *et al*., 1996). Species such as *O. chrysurus*, *L. triqueter*, *C. atringa* and *S. aurofrenatum*, all spent the largest proportion of their time approaching the trap mesh and were often observed making physical contact with the mesh. With the exception of *C. atringa* and *L. triqueter*, these species were also observed with scrapes on their snouts from butting into the trap mesh. While the existing research on alarm responses is limited (Smith, 1992), blood and other chemicals emitted by injured or wary fishes in traps may attract or repel conspecific or heterospecific species (Brown, 2003).

Evidence of conspecific attraction has been extensively examined elsewhere (Munro *et al*., 1971; Munro, 1974; Luckhurst & Ward, 1987), but the underlying mechanisms are not well documented. Visual cues such as fish shape and colour have been documented as important factors utilized by fishes for conspecific attraction and also aggression (Keenlyside, 1955; Blaxter, 1980). Recently, however, Ward *et al*. (2008) suggested that chemical cues may be more important than visual cues as they enable fishes to identify conspecifics from heterospecifics with a high degree of specificity and thus influence social behaviour. Research on sound detection and auditory cues in fishes suggests that sound is an important mechanism in which they can detect the location of conspecifics and heterospecifics (Popper & Lu, 2000). All these sensory cues probably influence the behaviour of fishes inside and outside the trap perhaps contributing to the behavioural overlap and attraction of both conspecifics (*L. triqueter*, *C. atringa* and *A. chirurgus*) and predators such as the *L. cyantopterus* and *C. hippos*.

A successful sustainable fishery is dependent not only on the function of the gear, but also on the basic knowledge of the behaviour of the fishes being caught. With the increase in the availability of low-cost video equipment, scientists, fishermen and managers can gain a greater understanding of how fishes are affected by fishing gear. The results presented from this study provided insight into the behaviour of fishes in response to traps in the U.S. Virgin Islands using underwater video surveillance techniques. Although significant differences were not detected in the amount of time species allocated to different behaviours, certain behaviours did appear to be species-specific thus offering a baseline for future work. Applying the insights gained from this study, future research should evaluate the behavioural responses of fishes to traps on the scale of the fishery in multiple habitats, depths, soak times and trap designs utilized by fishermen. The understanding of fish behaviour in traps has important management implications for trap fisheries and is particularly relevant to the use of escape gaps (Munro *et al*., 2003; Johnson, 2010) as an approach for managing multispecies trap fisheries, thus maximizing the efficiency and sustainability of such fishing gears.

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SUPPORTING INFORMATION

Supporting Information may be found in the online version of this paper:

Table SI. The distinct, identifiable fish behaviours observed from species caught inside the trap

Movie SI. Video footage displaying examples of behavioural interactions and activities that may occur both inside and outside of a fish trap. Trapped fishes can be seen approaching the mesh, particularly in the corners.

Movie SII. Video footage displaying an example of fish entrances and exits through the trap entrance funnel.

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